



Starter File: ReactionFurnance Starter.hsc
Sruf_PlantData_HistorianNew.xlsm

Solution file: ReactionFurnace.hscz

Version: V12.1

The AI Training feature in Aspen HYSYS enables you to use data and machine learning techniques to improve the accuracy of one or more models; used to compensate for deficiencies in the model. A neural network embedded within the model is trained using data to predict one or more model parameters as a function of feed and operating conditions.

Examples of such parameters include:

- Efficiencies in distillation columns
- Rate constants or reaction rates
- Heat transfer coefficients or heat transfer rates
- Mass transfer rates
- Pipe friction factors
- Pump and compressor efficiencies

In this document, we highlight the different steps required to create such a First Principles Driven Hybrid Model to accurately capture these effects.

- Step 1 – Import Data
- Step 2 – Condition Data
- Step 3 – Evaluate Model
- Step 4 – Build, Train and Validate Model
- Step 5 – Deploy Model

General Workflow

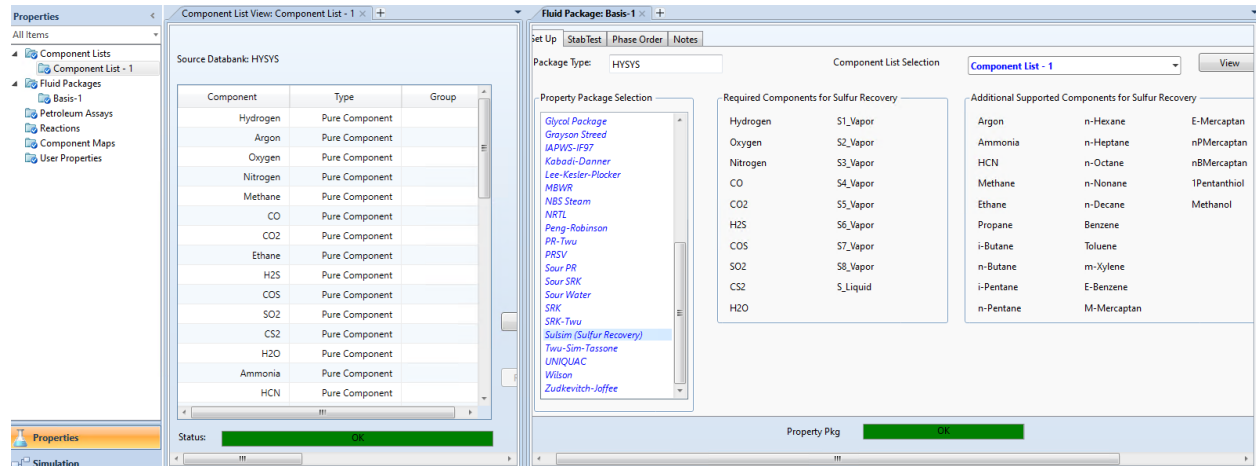


Reaction Furnace H₂S Oxidation.

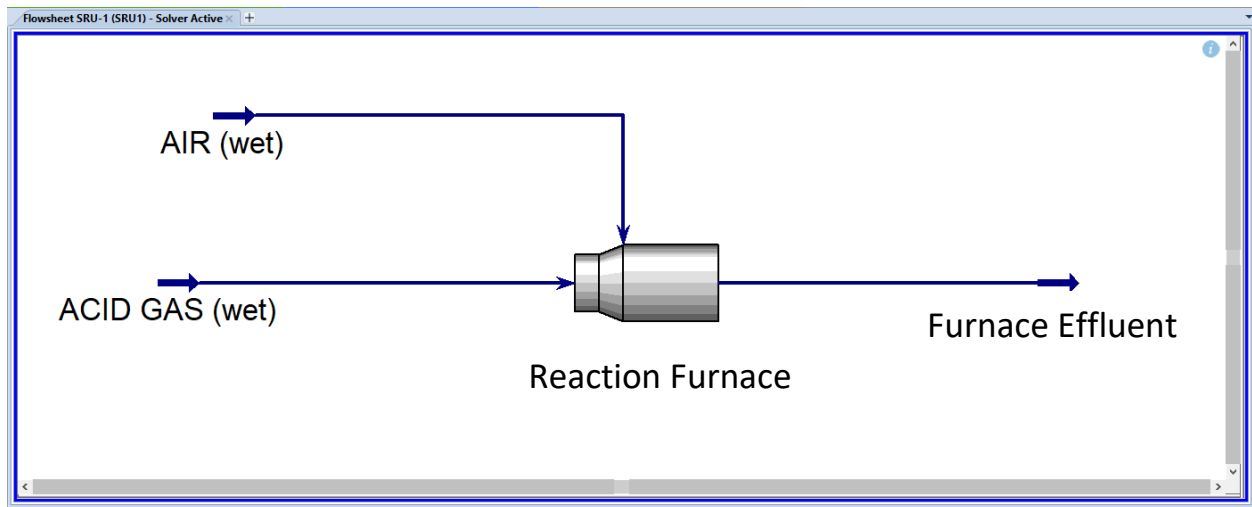
The objective of this example is to ‘train’ a model with plant data using First Principles-Driven Hybrid Models in Aspen HYSYS to help us optimize the oxidation of H₂S in the reaction furnace at a high temperature.

1. Start by opening **ReactionFurnace Starter.hsc**

The starter file has been defined for components supported by Sulsim (Sulfur Recovery) property package.



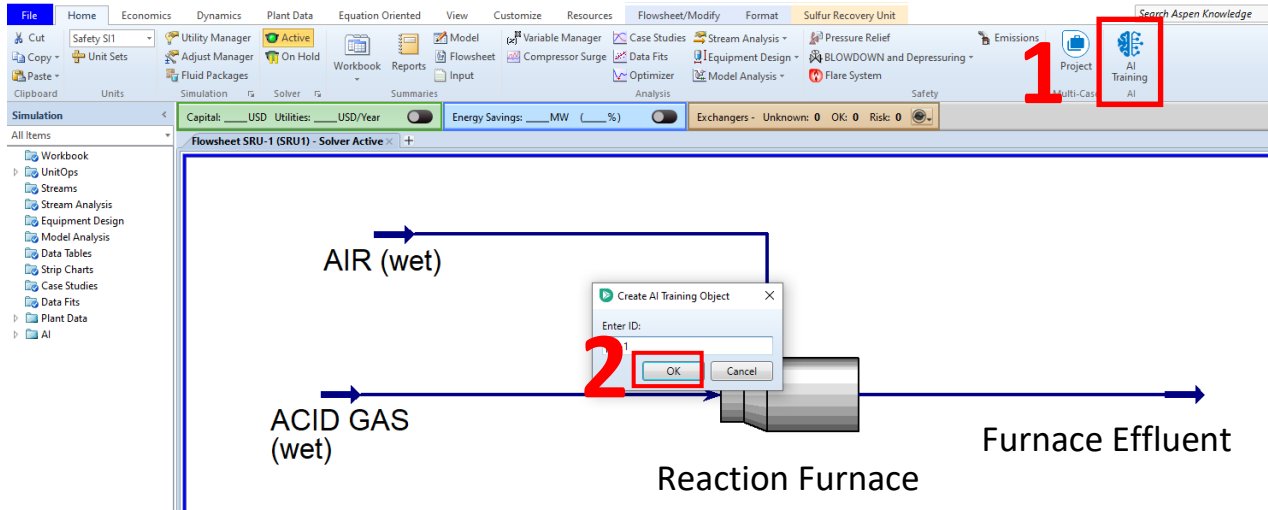
The simulation environment has already been defined using the SRU template with a Reaction furnace block specified and solved.



To train the model follow the listed steps:

Step 1 – Import Data.

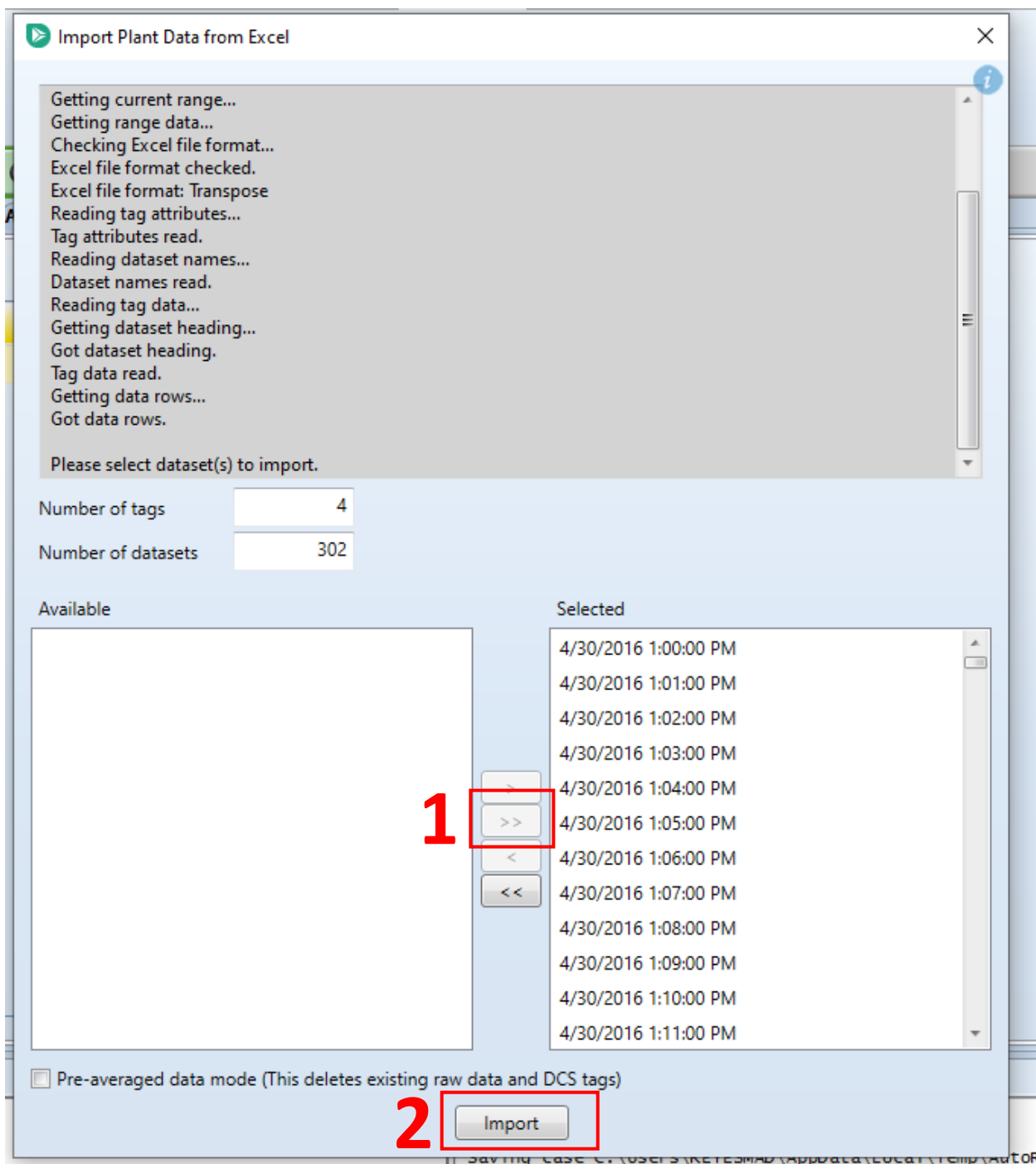
2. Once in the Sub-Flowsheet Environment, click the **AI Training** icon, located at **Home tab | AI**. Click **OK** for the **AI Training Object** manager; the default name for our model TO-1.



3. The AI Training tab will enable. Click the **Get Plant Data** button and, from the options, select **import from excel**, browse for the **Sruf_PlantData_HistorianNew.xlsx** file and click open.

Note: Remember that the organization of this tab starts on the left side uploading the data and finishes with the deployment of your AI model at the right end.

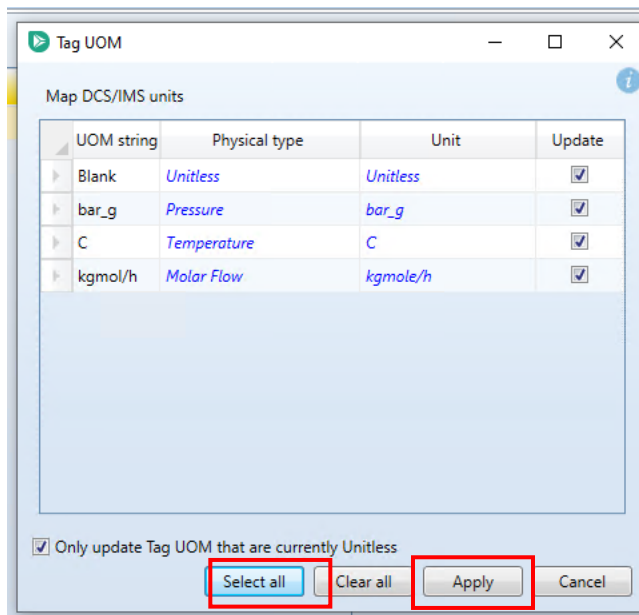
4. The **Import Plant Data manager** will display, and information will be loaded, it will show 2 tags and 302 data sets. Select all of them, click the >> button to transfer all 302 datasets; then click **Import**.



Note: once all data is imported the Import manager will display the message **Plant Data Successfully Imported.**

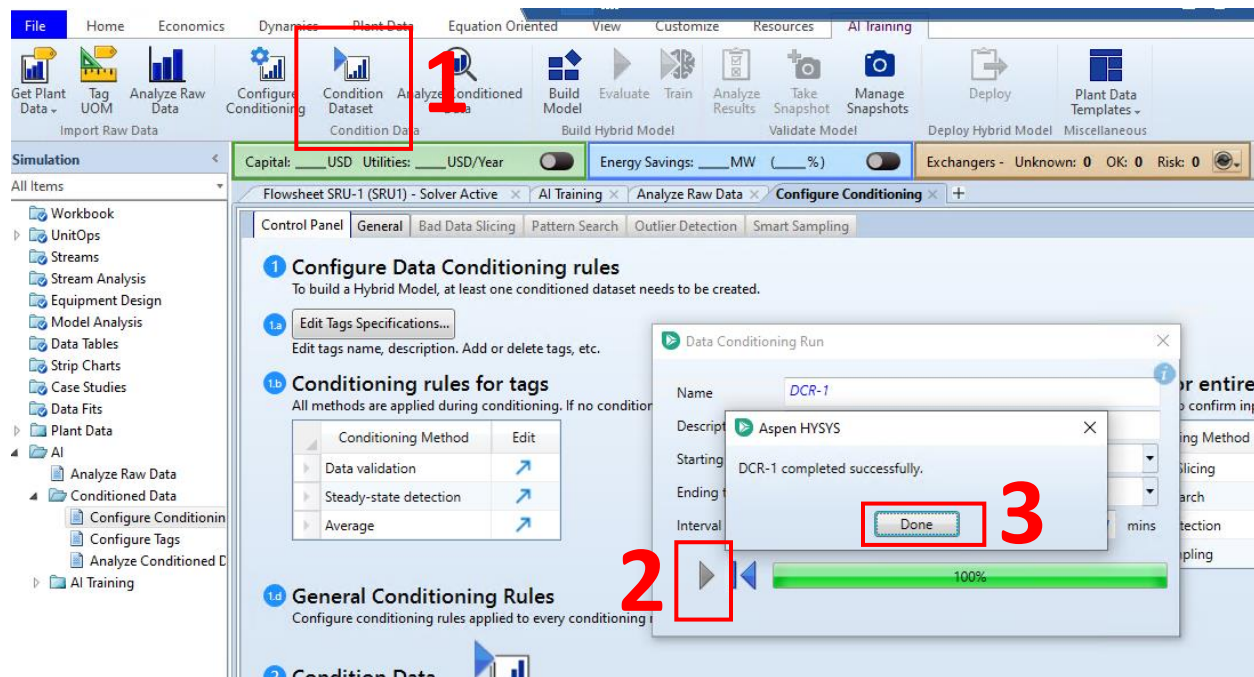


5. Close the Import Plant Data from Excel, and Map the DCS/IMS unit now shows, use bar_g for pressure, C for temperature and kgmole/h for Molar Flow, click **Select** all then **Apply**.



Step 2 – Conditioning Data.

6. **Configure Conditioning Dataset** will not be needed for this case. Click the **Condition dataset**; the **Data Conditioning Run manager** will appear, click **Run** then **Done**

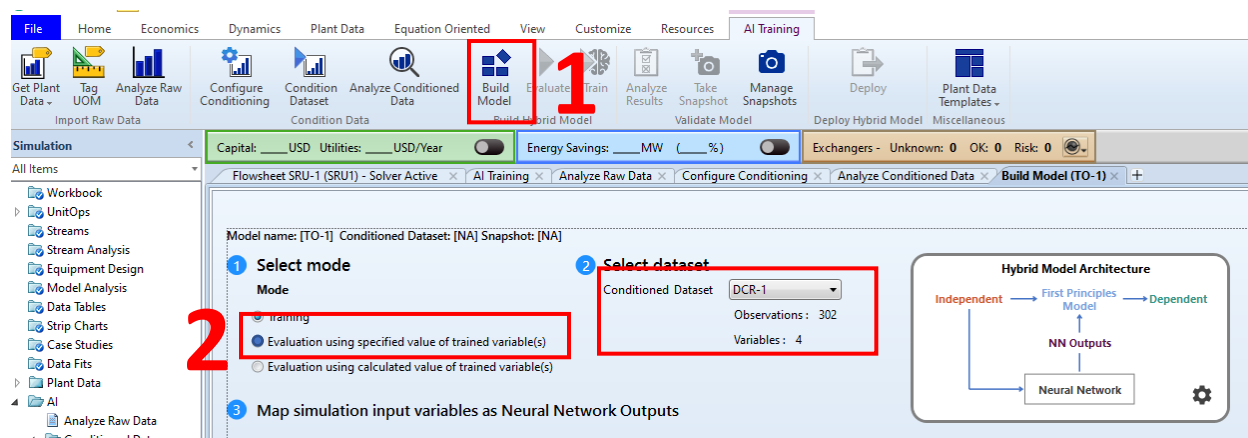




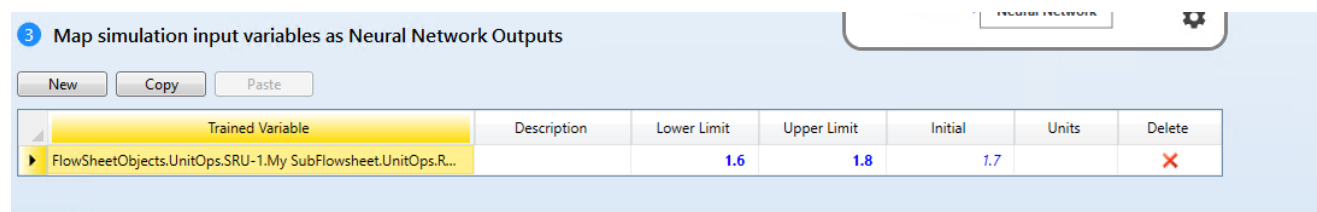
Note: Remember that if you detect any outlier, missing data, bad oscillating data point or points, in **Configure Conditioning** you can define the rules for the tags and data sets and set general conditions.

Step 3 – Evaluate Model.

- Click the **Build Model** button at the AI Training ribbon and select **Evaluate Using Specified Value of Trained Variable(s)** under **Mode**, **DCR-1** will be set as the conditioned set with 302 observations and 4 variables.



- Now map the simulation input variables. For input variables as Neural Network Outputs you can browse or copy and paste the variable **Air Stream to Inlet Gas Molar Flow Rate Ratio**. Define a **Lower Limit** of 1.6, **Upper Limit** of 1.8 and **Initial Value** of 1.7.



- Map the tags as independent or dependent variables, you can browse for them or copy and paste as follows:

Description	Stream/UnitOp.	Variable Name	Dependent/Independent
Acid Gas Pressure	Acid Gas (wet)	Phase- Pressure Overall	Independent
Effluent H2S	Furnace Effluent	Molar Flow- Overall H2S	Dependent
Acid Gas Temperature	Acid Gas (wet)	Phase-Temperature Overall	Independent
Acid Gas H2S	Acid Gas (wet)	Molar Flow- Overall H2S	Independent



4 Map Tags to simulation variables or to user-defined custom variables
Assign each Tag as Independent or Dependent (simulation inputs = Independent, simulation outputs = Dependent)

☒ Include in Model	Tag	Description	Units	Variable Type	Variable Name	Independent	Dependent
☒	A1002.PV	Acid Gas Pressure	bar_g	Simulation	FlowSheetObjects.UnitOps.SRU-1.My SubFl...	☒	☐
☒	A1011.PV	Effluent H2S	kgmole/h	Simulation	FlowSheetObjects.UnitOps.SRU-1.My SubFl...	☐	☒
☒	F1001.PV	Acid Gas Temperature	C	Simulation	FlowSheetObjects.UnitOps.SRU-1.My SubFl...	☒	☐
☒	H1003.PV	Acid Gas H2S	kgmole/h	Simulation	FlowSheetObjects.UnitOps.SRU-1.My SubFl...	☒	☐

10. Click Evaluate and observe the results.

Step 4 – Build, Train and Validate Model.

11. Go back to **Build Model** and Select **Training** mode.

The screenshot shows the Aspen Tech software interface. The 'Build Model' tab is active. The 'Build Model' button is highlighted with a red box and a red number '1'. The 'Training' mode is selected under 'Select mode' with a red box and a red number '2'. The 'Hybrid Model Architecture' diagram is visible on the right.

12. Click **Neural Network** button under Hybrid Model Architecture manager and for **Training set fraction** enter 0.8, **Hidden layers** 1, **Training Epochs** 100. 8 **Neurons** and select Sigmoid for the **activation** function. Click **Train**.



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Neural Network Configuration

Hybrid Model Architecture

Independent → First Principles Model → Dependent

NN Outputs

Neural Network

Training set fraction: 0.8 Hidden Layers: 1 Training Epochs: 100

Hidden Layer	Neurons	Activation	Activation Parameter
1	8	Selu	0.1

Output Layer

Predicted Parameter	Activation	Activation Parameter	Lower
FlowSheetObjects.UnitOps.SRU-1.My SubF...	Sigmoid	0.1	

Initialization

☒ Cold start

☐ Reinitialize from previously converged run

13. Results for the AI will display showing approximately 0.996 accuracy in the Dependent Variables Fit Map plots with the relationships of variables and the H2S Effluent.



Step 5 – Deploy Model.

14. Model is ready to be deployed, click **Deploy**, TO-1 AI model will not be attached to the Reaction Furnace Block, with the trained variables.



AI Calculator: TO-1

Connections Parameters About

AI Calculator Name **TO-1**

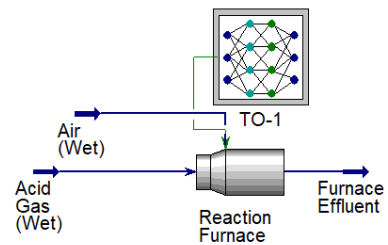
Neural Network Inputs

Tag	Object	Variable Description	Value
F1001.PV	Acid Gas (Wet)	Phase - Temperature (Overall)	35.00
H1003.PV	Acid Gas (Wet)	Overall and Phase Comp Molar Flow (Overall-H2S)	71.93

Neural Network Outputs

Tag	Object	Variable Description	Value
NNOutput-1	Reaction Furnace	Air Stream to Inlet Gas Molar Flow Rate Ratio	1.708

Delete OK



Activate Windows